# hdoc day 36 by trishil chauhan

## Aim

Write a menu-driven C program using separate functions to count characters without a built-in length function, display each character on a new line, count vowels and consonants, and exit.

## C Program

```
#include <stdio.h>
#include <ctype.h>

#define MAX 1000

int countCharacters(const char str[]) {
    int count = 0;
    while (str[count] != '\0') {
        count++;
    }
    return count;
}

void printCharacters(const char str[]) {
    int i;
    for (i = 0; str[i] != '\0'; i++) {
        printf("%c\n", str[i]);
    }
}

void countVowelsConsonants(const char str[]) {
    int vowels = 0, consonants = 0, i;
    char ch;
    for (i = 0; str[i] != '\0'; i++) {
        ch = (char)tolower((unsigned char)str[i]);
        if (ch >= 'a' && ch <= 'z') {
            if (ch == 'a' || ch == 'e' || ch == 'i' ||
                ch == 'o' || ch == 'u') {
                vowels++;
            } else {
                consonants++;
            }
        }
    }
    printf("Vowels: %d\n", vowels);
    printf("Consonants: %d\n", consonants);
}

int main(void) {
    char str[MAX];
    int choice;

    printf("Enter a string: ");
    if (fgets(str, sizeof(str), stdin) == NULL) {
        return 1;
    }
    /* Remove the newline added by fgets, if present. */
    int len = countCharacters(str);
    if (len > 0 && str[len - 1] == '\n') {
        str[len - 1] = '\0';
    }

    do {
```

```
        printf("\n--- STRING OPERATIONS ---\n");
        printf("1. Count characters\n");
        printf("2. Print each character on a new line\n");
        printf("3. Count vowels and consonants\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");

        if (scanf("%d", &choice) != 1) {
            printf("Invalid input.\n");
            return 1;
        }

        switch (choice) {
            case 1:
                printf("Character count: %d\n",
                       countCharacters(str));
                break;
            case 2:
                printCharacters(str);
                break;
            case 3:
                countVowelsConsonants(str);
                break;
            case 4:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice. Try again.\n");
        }
    } while (choice != 4);

    return 0;
}
```

## Sample Output

```
Enter a string: Hello World

--- STRING OPERATIONS ---
Enter your choice: 1
Character count: 11

Enter your choice: 3
Vowels: 3
Consonants: 7

Enter your choice: 2
H
e
l
l
o

W
o
r
l
d

Enter your choice: 4
Exiting program.
```

## Result

The program successfully performs all three string operations through separate user-defined functions. Spaces count as characters but not as vowels or consonants.